Xueyan Zhang

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Research Interests

- **Understanding LLMs from the inside**: how information flows across layers, how steering a few activations changes behavior, and where visual information enters the language stream.
- **Using what we find to improve models**: post-training and data selection guided by internal signals, parameter-efficient fine-tuning, optimizers, and model merging.
- **Making them fast**: KV cache, speculative decoding, and hardware-aware compilation, from kernels to serving.
- Also drawn to latent-space communication between agents, and LLMs as verifiable compilers.

Education

- **University of Waterloo**, *Master of Math in Computer Science* — Waterloo, Ontario
  Research Stream in Software Engineering; Avg. 93/100 — *Jan. 2021 – Dec. 2022*
  Graduate Excellence Award in Computer Science; International Master's Award of Excellence
- **McMaster University**, *Bachelor of Engineering in Mechatronics Co-op (with Distinction)* — Hamilton, Ontario
  GPA: 11.5/12 — *Sept. 2015 – Apr. 2020*
  Dean's Honour List; Senate Scholarship; Woo Family International Scholarship

Publications

- **UORA: Uniform Orthogonal Reinitialization Adaptation in PEFT** *ACL 2025*
  - **First author.** A novel interpolation-based reinitialization mechanism for parameter-efficient fine-tuning (**PEFT**), reducing trainable parameters by up to **15x** vs. LoRA while staying competitive across NLU, NLG, instruction-tuning, and ViT.
- **Syntactic Prediction through Reinforcement Learning** *EMNLP 2026*
  - **Corresponding author.** Introduced **Syntax-RL**: cold-start SFT, then **GRPO** on structure-level rewards, breaking the SFT plateau; SOTA on constituency parsing (**96.51** F1 on PTB, above the best prior neural parser) and CCG supertagging.
- **Tiny Budgets, Big Gains: Parameter Placement Strategy in PEFT** *EMNLP 2025*
  - Introduced **FoRA-UA**, a sparse Fourier low-rank method with a universal shared adaptor, reaching SOTA with only **1–5%** of LoRA's parameters.
- **Large Language Model is not a (Multilingual) Compositional Relation Reasoner** *COLM 2024*
  - **Co-first author.** Created and open-sourced the **MCR Benchmark**, a multilingual evaluation of LLM compositional reasoning across six relation categories in five languages.
- **On the Caching Schemes to Speed Up Program Reduction** *TOSEM 2024*
  - Master's research. Proposed Refreshable Compact Caching (**RCC**), cutting **64%** redundant queries, accelerating reduction by up to **43%**, and shrinking memory **99.97%** (100+ GB → 6 MB).
- **e2a: A Secure Agent-Native Email Gateway with Phishing and Prompt-Injection Defense** *EMNLP 2026 Demo*
  - Open-source email gateway for AI agents; its benchmark shows email structure defeats content-only injection detectors.
- **The More You Say, the More You Risk: Ethical Concerns in LLM Reasoning** *WWW 2026 Workshop*
  - Showed that CoT, Self-Reflection, and Multi-Agent Debate erode alignment on stereotype benchmarks across 6 LLMs and 4 languages; Multi-Agent Debate cut reasoning safety by **34.4%** on average.

Preprints

- **How Much Operator Fusion Is Left? A Compiler-Side Bound & Learned Policy** *Preprint*
  - **First author.** Established a compiler-side headroom bound (fusible work only ∼**24%** of GPU time), then trained a learned graph-level fusion policy (SFT+GRPO) for PyTorch Inductor that fuses **39%** more nodes/graph at **28.5%** fewer kernels, holding runtime parity with TorchInductor and **+7.6%** on untuned shapes. *Under review.*

- **HR-Merge: Model Merging with Headroom-Weighted Residual Refinement** *Preprint*
  - **Co-first author.** A model-merging method using headroom-weighted residual refinement to combine multiple fine-tuned models. *Under review.*
- **TARE-Muon: Timescale-Aware Row-wise Energy Redistribution** *Preprint*
  - A Muon-family optimizer that redistributes row-wise update energy using long- and short-term momentum.
- **ATPO: Adaptive Token Aggregation for Policy Optimization** *Preprint*
  - Adaptive token aggregation for policy optimization in RL fine-tuning of LLMs.
- **IntentEval: Do LLMs Answer What Users Actually Mean?** *Preprint*
  - A benchmark testing whether LLMs answer what users mean under vague or false-premise questions.
- **Can Language Model Understand Word Semantics as A Chatbot?** *Preprint*
  - **Co-first author.** Quantified internal-vs-external knowledge mismatch for word semantics across Encoder, Decoder, and Encoder-Decoder LMs; probing internal states recovers more accurate semantics than querying outputs. *arXiv preprint.*

Experience

- **Toronto Heterogeneous Compiler Lab** Toronto, Ontario
  Senior NPU Compiler Engineer *Sept. 2022 – June 2026*
  - **NPU Compiler Optimization for LLMs**:
    - Spearheaded compiler optimizations in LLVM for featured networks, like **Flash Attention** and RMS norm, achieving **over 30%** improvement in operator performance and a 10% reduction in compiled binary size.
    - Served as the technical focal point for a small-sized team, leading communication and aligning technical progress with international counterparts.
    - Designed a suite of custom compiler passes: enhanced recognition and transformation of zero-overhead hardware loops, plus ahead-of-time constant propagation, prefetching, custom loop fusion, and peephole optimizations.
    - Collaborated with teammates on loop and memory optimizations in the NPU backend: **loop tiling**, loop rotation, **vectorization** (SIMD and SIMT), and **software pipelining with double buffering** to overlap data movement with compute and remove pipeline bubbles.
    - Worked on **automatic synchronization**: analyzed read/write dependences between memory accesses to remove redundant user-written **memory barriers** and insert missing ones automatically.
  - **Unified Test Case Reduction Framework**:
    - Architected an award-winning test case reduction framework that industrializes my **program-reduction research**: unified 5 open-source algorithms and pioneered a novel, LLM-based approach to auto-generate test properties.
    - This service is now used by **200+** developers and has boosted debugging efficiency by 50%, automatically minimizing a 10k-line test case to its 100-line root cause in under 30 minutes.
  - **AI-powered Code Review Service**:
    - Built an AI-powered code review agent on **Jenkins**, integrating **Deepseek** as the LLM backend and **OpenCode** as the agent framework, with multi-agent collaboration for automated merge request analysis.
    - Configured GitLab CI pipelines and provided glab CLI tooling for MR view, diff, and API operations, enabling fully automated review feedback on every merge request.
    - Deployed Deepseek V4 Flash with 1M context window via **vllm** on two-node NVIDIA GB10.
  - **Data Movement Interface Re-architecture**: Redesigned data movement interface, introducing LLVM intrinsics and MIR pseudos that eliminated error-prone code and simplified analysis for downstream passes.
  - **General Matrix Multiplication in MLIR**: Engineered a vectorization feature in MLIR to pattern match, transform, and optimize user codes to fully utilize the matrix computing unit. Supported double buffering.
  - **Student Intern Mentorship**: Mentored multiple software engineering interns on compiler projects, leading to the successful integration of their features into the production codebase and a final results presentation to headquarters.
- **University of Toronto**, *Volunteer Research Assistant (Part-time)* Toronto, Ontario
  LLM reasoning and parameter-efficient fine-tuning with Prof. Gerald Penn and Dr. Jinman Zhao. *May 2023 – Present*
- **UWaterloo CS136**, *Instructional Apprentice* Waterloo, Ontario
  Clarified algorithms and data-structure concepts in weekly office hours. *Sept. 2021 – May 2022*
- **UWaterloo SWAG Lab**, *Undergraduate Research Assistant* Waterloo, Ontario
  Engineered research benchmarking automation with tracking, quantifying, and visualizing. *May 2020 – Dec. 2020*

Projects

- **OpenMarkdown.dev**, *Local-first Markdown editor with agent co-editing* *Jul. 2026*
  - Built a local-first desktop editor with a **Rust** core that exposes a **Model Context Protocol (MCP)** interface for real-time AI-agent co-editing; **reached #5 on Product Hunt** at launch.
  - Designed a conflict-resolution scheme that merges concurrent human and agent edits without data loss.
- **OpenTikZ.org**, *Open-source TikZ library for AI-assisted academic diagrams* *May 2026*
  - Designed and shipped a community TikZ resource library, the "Flaticon for academic TikZ", delivering **100+** reusable assets for system, neural-network, and pipeline figures; **240+** GitHub stars.
  - Shipped a **Claude Code** skill, enabling an AI agent to discover, edit, and recompile figures from natural-language requests so researchers produce publication-ready diagrams without hand-writing TikZ.
- **iMarkAi.com**, *Agentic bookmark with semantic search & web interactions* *Nov. 2025*
  - Designed, implemented, and published a commercial-grade bookmark extension on Chrome Web Store.
  - Architected an intelligent search platform using **Pinecone** vector database and Google Gemini AI, achieving sub-second semantic retrieval across unstructured web data with 95%+ precision.
  - Implemented end-to-end **Stripe** billing for subscriptions and usage-based payments.
- **Tarot: Starry Guidance**, *End-to-end mobile app development* *Sept. 2025*
  - Led the end-to-end, full-stack development on the WeChat mini-program platform; launched within 1 month.
  - Utilized web scraping for data; implemented HTML/CSS for UI, a cloud database, and an integrated LLM for personalized interpretations of drawn cards.
  - Conducted market research and competitor analysis; **reached 500+ users in first 3 months**.

Technical Skills

- **Programming**: C/C++, Python, Bash, Java/Kotlin, Unix/Linux, git Git
- **Toolbox**: Huggingface, Pytorch, LLVM, MLIR, Bazel, Docker, Claude Code/API/SDK, NVIDIA Spark

Service

- Reviewer, ICLR 2025; AAAI 2027; ACL ARR 2026
- Delivered Prompt Engineering, Reinforcement Learning, and Fine-tuning Crash Courses